Macroeconomics I — MPE Insper

Practice list: The New Keynesian AS–AD model (Prova 01 remediation, Q3)

Instructions. Multiple choice: mark the single correct option (no penalty for wrong answers). *True/False*: every 2 wrong items cancel 1 right one (score = $\max\{0, \text{right} - \text{wrong}/2\}\times$ item value); blank items do not count. Discursive questions: show every step. Parameters are symbols with the stated admissible ranges; answers are closed forms, conditions, signs and limits.

Notation (as in Prova 01; Benigno 2015). Lower-case letters are logs (deviations from the steady state); bars denote the long run; $\Omega \equiv \sigma^{-1} + \eta$; $\rho \equiv -\ln\beta$, so $-\sigma\ln\beta = \sigma\rho$.

$$y = \bar{y} + (g - \bar{g}) - \sigma[i - (\bar{p} - p)] - \sigma\ln\beta, \tag{AD}$$

$$p - p^e = \kappa(y - y_n), \tag{AS}$$

$$y_n = \tfrac{1+\eta}{\Omega}\, a + \tfrac{\sigma^{-1}}{\Omega}\, g - \tfrac{1}{\Omega}\,\mu, \tag{YN}$$

$$\bar{y} = \tfrac{1+\eta}{\Omega}\, \bar{a} + \tfrac{\sigma^{-1}}{\Omega}\, \bar{g} - \tfrac{1}{\Omega}\,\bar{\mu}, \tag{YB}$$

$$y_e = \tfrac{1+\eta}{\Omega}\, a + \tfrac{\sigma^{-1}}{\Omega}\, g, \tag{YE}$$

$$L = \tfrac{1}{2}(y - y_e)^2 + \tfrac{\phi}{2\kappa}(p - p^e)^2, \qquad \phi \geq 0 \qquad \text{(loss of the central bank)}. \tag{LF}$$

a, μ, g are current productivity, (log) markup and public spending; $\bar{a}, \bar{\mu}, \bar{g}$ their long-run values. Parameters: $\sigma, \eta, \kappa > 0$. Unless stated, the economy starts at an efficient steady state ($y = y_n = y_e = \bar{y}$, $p = p^e = \bar{p}$, $g = \bar{g}$; a subsidy offsets the steady-state markup, so $\mu, \bar{\mu}$ are deviations from it), the bank sets i, and p^e and $\bar{p}$ do not move. "Shock" means a change $d(\cdot)$ from that point.

Block 1 — Exam level: Benigno's own experiments, Lista 7 and Carlin–Soskice

(Benigno §§6, 7, 11 • 40 points)

Question 1. (10 points) *Source: built on Benigno, P. (2015), "New-Keynesian Economics: An AS–AD View", §6.1–§6.2, Figs. 6–7, p. 512 — the article has no exercises; the statement below is ours. The article treats a productivity* gain*; here the sign is reversed.* Redo the two experiments of those sections for a productivity *fall*, $da < 0$.

What the article states (p. 512): "First, we analyze the case in which the economy undergoes a temporary productivity gain, meaning that productivity rises in the short run but does not vary in the long run." (§6.1) and "Fig. 7 analyzes the case of a permanent productivity shock." (§6.2)

a) (4 points) The fall is temporary ($d\bar{a} = 0$) and i stays at its initial level. Which curve moves, in which direction and why? Find dy, dp, $d(y - y_n)$ and $d(y - y_e)$ in closed form, with signs.

Solution:

b) (3 points) Find the change in i that keeps $p = p^e$. Is it a cut or a rise? What does it cost in terms of $y - y_e$?

Solution:

c) (3 points) Now the fall is permanent ($d\bar{a} = da$). Answer (a) and (b) again and explain the difference.

Solution:

Question 2. (10 points) *Source: built on Benigno, P. (2015), "New-Keynesian Economics: An AS–AD View", §6.3, Fig. 8, p. 513 — the article has no exercises; the statement below is ours. Item (a) redoes the article's experiment; items (b)–(c), the markup analogue, are added by this list.*
What the article states (p. 513): "Consider now the case of an expected increase in long-run productivity. [...] AS does not move, since there is no change in current productivity."

a) (4 points) Redo the optimism experiment of §6.3 in the exam notation: $d\bar{a} > 0$, $da = 0$, i fixed. Curve, dy, dp, both gaps, the di that keeps $p = p^e$ and its cost.

Solution:

b) (4 points) Instead of $\bar{a}$, the news is that the long-run markup will *fall*, $d\bar{\mu} < 0$ (a competition reform that takes effect next period); μ does not change. Answer (a) for this shock and draw the AS–AD diagram with E_0, the new equilibrium E', the gap $y - y_n$, and the AD after the policy of price stability.

Solution:

c) (2 points) In words: what changes if the news is a *rise* in $\bar{\mu}$ (Prova 01, Q3)? Does the answer depend on ϕ?

Solution:

Question 3. (10 points) *Source: MPE Macroeconomics I, Lista 7 (2026), Question 2, item (a) only — reproduced verbatim (its points label and item (b) omitted); built also on Benigno, P. (2015), "New-Keynesian Economics: An AS–AD View", §7, Fig. 9, pp. 513–514, which has no exercises.*
Lista 7, Question 2. Using separate AS–AD diagrams, analyze the following changes:

a) a temporary increase in firms' desired markups;

In each case, identify which curve shifts, indicate the direction of the shift, and describe the effects on output and the price level. Explain the economic intuition behind the movement of the curve and the resulting equilibrium.
What Benigno states (p. 513): "This section considers only temporary mark-up shocks, leaving other analyses to the reader."
Added parts (not in Lista 7), with the sign changed: the desired markup *falls* for one period only, $d\mu < 0$ (a temporary payroll-tax cut, or a fall in the oil price); $d\bar{\mu} = 0$. Items (a)–(c) below are this list's.

a) (3 points) With i fixed: which curve moves and where; dy, dp, $d(y - y_n)$, $d(y - y_e)$ with signs.
Solution:

b) (3 points) Find the di that keeps $p = p^e$ and the di that keeps $y = y_e$. Report $y - y_e$ in the first case and $p - p^e$ in the second. Is there a trade-off?
Solution:

c) (4 points) The bank minimizes L. Find dy^*, dp^* and di^*, and the condition on ϕ under which it cuts.

Solution:

Question 4. (10 points) *Optimal policy. Sources: MPE Macroeconomics I, Lista 7 (2026), Question 4 — reproduced verbatim (its points label omitted); Carlin, W. and Soskice, D. (2024),* Macroeconomics: Institutions, Instability, and Inequality*, §3.6.1 checklist questions 4 and 5, p. 156 — reproduced verbatim; built also on Benigno, P. (2015), "New-Keynesian Economics: An AS–AD View", §11, eqs. (33)–(35), Fig. 14, pp. 521–523, which has no exercises.*

Lista 7, Question 4. The microeconomic foundations of the New Keynesian model make it possible to evaluate monetary policy in terms of household welfare.

Consider the following loss function in conjunction with the AS-AD dynamics seen in Benigno (2015)

$$\min_{y,p} \mathcal{L} = \phi_y(y - y^*)^2 + \phi_p(p - p^e)^2, \qquad \phi_y, \phi_p > 0,$$

s.t:

$$p - p^e = \kappa(y - y_n).$$

Interpret the shape of the loss function and analyse what it implies for the behavior of a central bank. Also, explain why this problem is subjected to the AS condition and what does that mean in terms of economic policy

Carlin and Soskice, checklist questions 4 and 5.

4. Explain what is meant by the central bank's loss function. How are the central bank's preferences reflected in the loss function? Draw the loss 'circles' for the cases where
   (a) $\beta = 1$;
   (b) $\beta < 1$;
   (c) $\beta > 1$.

   In which of the three cases will the central bank reduce inflation back to target quickest after an inflation shock? Is there any downside to adopting this policy stance?

5. Assume that $\alpha = \beta = 1$, derive the MR curve graphically using the tangencies between the loss circles and the Phillips curves. With reference to the diagrams, explain the effect of the following (in each case, assume all other parameters are held constant):
   (a) An increase in the slope of the Phillips curve, α
   (b) An increase in central bank's inflation aversion, β.

What Benigno states (p. 522): "This rule is robust to all types of shock and does not depend on their properties."

Added parts (not in the sources): items (a)–(c) below are this list's. They ask the questions above in Benigno's static model and with the loss L of the cover: ϕ plays the role of Carlin and Soskice's β, κ of their α, the IT line of their MR curve.

a) (4 points) Minimize L subject to the AS. Derive the targeting rule, explain why the AS (and not the AD) is the constraint, and describe the loss contours and the IT line. Show that the Lista 7 loss $\phi_y(y-y^*)^2+\phi_p(p-p^e)^2$ gives the same rule for suitable weights.

Solution:

b) (3 points) Let $W \equiv y_e - y_n > 0$. Find $y^* - y_e$ and $p^* - p^e$, sign their derivatives in ϕ and κ, and give the limits $\phi \to \infty$ and $\phi = 0$. Relate the answer to the two Carlin–Soskice questions.

Solution:

c) (3 points) For a shock with changes dy_n, dy_e, $d\bar{y}$ (g, $\bar{p}$, p^e fixed), find di^* and prove that it depends on ϕ if and only if $dy_n \neq dy_e$. Apply the result to news of a higher $\bar{\mu}$.

Solution:

Block 2 — Harder and conceptual: which bar moves which curve, signs, divine coincidence, zero bound

(Benigno §§4–11 • 38 points)

Question 5. (10 points) *Which bar moves which curve.* Consider a small rise, one at a time, in each of $a, \bar{a}, \mu, \bar{\mu}, g, \bar{g}, \bar{p}, p^e$.

a) (6 points) Build a table with, for each variable: the curve that moves and its shift (horizontal for AD and AS; vertical for p^e); whether y_e moves; whether there is a trade-off; and the di (sign and closed form) that keeps $y = y_n$ and $p = p^e$.

Solution:

b) (2 points) In words: state the rule that tells you which curve a variable moves.

Solution:

c) (2 points) In words: why is μ the only row with a trade-off?

Solution:

Question 6. (8 points) *Diagnose the answer (words only, no algebra).* News arrives that the long-run markup will rise ($d\bar{\mu} > 0$; current μ unchanged). A student writes four claims. For each, say whether it is right and, if not, correct it in words.

(1) "Firms will charge higher markups, so the AS shifts left."

(2) "Along the AD, prices go up and output goes down."

(3) "To keep prices stable the bank must raise i, accepting a deeper recession as the price of stable prices."

(4) "The optimal response depends on whether ϕ is larger than σ, i.e. on whether the IT line is

flatter than the AD."

Solution:

Question 7. (9 points) *When does the divine coincidence fail?* True or false; justify each in words (1.5 points each).

a) If nominal wages were preset as well as prices, a productivity shock could create a trade-off between price stability and the efficient output gap.

b) A temporary rise in g forces the bank to choose between stable prices and a zero efficient gap.

c) Around a steady state with a large markup distortion, y_e need not be the welfare-relevant target, and the results for markup shocks are the ones most at risk.

d) A temporary rise in the consumption tax is a pure demand shock, so it creates no trade-off.

e) After bad news about $\bar{a}$, if the zero bound binds, the bank faces a trade-off between price stability and the efficient gap.

f) A dovish bank ($\phi < \sigma$) answers news of a higher $\bar{\mu}$ with a larger cut than a hawkish bank.

Question 8. (11 points) *The zero bound after $\bar{\mu}$ news.* Start from the steady state with $i = \rho > 0$. News arrives that the long-run markup will rise, $d\bar{\mu} > 0$; current μ is unchanged and the bank cannot set $i < 0$.

a) (3 points) Find the rate that keeps $p = p^e$ and the size $d\bar{\mu}^*$ above which the zero bound binds. Interpret with the natural rate.

Solution:

b) (3 points) Let $d\bar{\mu} > d\bar{\mu}^*$ and $i = 0$. Find dy, dp, $d(y - y_n)$, $d(y - y_e)$. Is there a trade-off?

Solution:

c) (3 points) Find the commitment $d\bar{p}$ that restores $y = y_n$ and $p = p^e$ with $i = 0$. Explain the channel and why credibility matters.

Solution:

d) (2 points) Alternatively, find the temporary dg that restores $p = p^e$ at $i = 0$, and say why fiscal policy works here.

Solution:

Block 3 — Created items: two shocks at once, and the timing of fiscal news

(created • 22 points)

Question 9. (12 points) *Created.* An oil-price rise lifts the current markup, $d\mu > 0$, and in the same week a law is passed that raises a payroll tax from next period on, $d\bar{\mu} > 0$. The bank minimizes L with a general $\phi \geq 0$.

a) (3 points) With i fixed: which curves move? Find dy, dp, $d(y - y_n)$, $d(y - y_e)$. When do prices rise?

Solution:

b) (4 points) Find dy^*, dp^*, di^* under the optimal rule. Split di^* into a part due to $d\bar{\mu}$ and a part due to $d\mu$; show which depends on ϕ; sign $\partial di^*/\partial\phi$; give the limits $\phi \to \infty$ and $\phi = 0$, and the

condition for a cut.

Solution:

c) (2 points) Let the two shocks be equal, $d\bar{\mu} = d\mu$ (a permanent markup rise). Show that the optimal bank cuts for every ϕ, and explain why, although with i fixed prices do not move.

Solution:

d) (3 points) Draw the AS–AD diagram with $\sigma = 1$, $\eta = 0.5$ ($\Omega = 1.5$), $\kappa = 0.5$, $\phi = 3$, $d\mu = 1.5$, $d\bar{\mu} = 0.75$: AS, AS$'$, AD, AD$'$, the IT line, E_0, the no-policy point E', the optimum O with its AD, and both gaps at O. Give di^* and its two parts.

Solution:

Question 10. (10 points) *Created.* Public spending rises by $\Delta > 0$ in three ways: (i) temporarily, $dg = \Delta$, $d\bar{g} = 0$; (ii) permanently, $dg = d\bar{g} = \Delta$; (iii) as news only, $d\bar{g} = \Delta$, $dg = 0$.

a) (4 points) For each case: which curves move and by how much; dy, dp, $d(y - y_n)$ with i fixed; and $d(y - y_e)$.

Solution:

b) (3 points) Find the optimal di^* in each case and show that ϕ does not enter. Check that (i) and (iii) add up to (ii).

Solution:

c) (3 points) Draw the three diagrams for $\sigma = 1$, $\eta = \kappa = 0.5$, $\Delta = 1.5$. Then, in words: what is the output multiplier when the bank follows its optimal rule, and how does case (iii) relate to the $\bar{\mu}$ news of Prova 01?

Solution: